

A Comparative Analysis of Optimization and Reinforcement Learning Approaches for the Parking Assignment Problem

Moosa Ahmed, Bhavnoor Dhillon, Reilly Macdonald, Lucas Matheson

Subjects:

Artificial Intelligence, Algorithms, Decision Making, Mathematical Optimization, Reinforcement Learning, Stochastic Analysis, Markov Processes

Abstract

This paper investigates the parking assignment problem in a dynamic urban road network, where drivers must select parking locations based on travel time, cost, availability, and walking distance constraints. The problem is modeled using both a classical search-based approach and a reinforcement learning framework. The first method combines Dijkstra's algorithm for shortest-path computation with a hill-climbing optimization strategy to select parking locations. The second method formulates the problem as a Markov Decision Process and applies Q-learning to enable adaptive decision-making.

Simulation results across multiple scenarios show that both methods successfully identify feasible parking solutions under varying constraints. The optimization-based approach achieves lower cost in static conditions, while the Q-learning method demonstrates greater adaptability to changes in parking availability and walking constraints. These findings highlight the trade-off between efficiency and adaptability, suggesting that reinforcement learning is more suitable for dynamic environments.

1. Introduction

Artificial Intelligence (AI) plays an important role in solving real-world optimization problems in dynamic and uncertain environments. One such problem is urban parking assignment, where drivers must select a parking location while considering travel time, cost, availability, and walking distance. Efficient solutions to this problem

can improve traffic flow and support smart city applications.

Efficient parking assignment systems are important in modern urban environments and smart cities. Poor parking decisions can lead to increased traffic congestion, longer travel times, and higher fuel consumption. Intelligent parking systems can improve driver experience by providing better guidance to available parking spaces and reducing unnecessary searching. Applications of such systems include navigation tools, smart city infrastructure, and real-time traffic management platforms.

In this project, we study the parking assignment problem in a dynamic road network with time-varying traffic conditions and changing parking availability. Traditional approaches model this problem as a graph-based search task, where nodes represent locations and edges represent possible transitions. While these methods can compute efficient routes, they are limited in their ability to adapt to changing conditions.

To address this limitation, we compare two solution approaches. First, we implement a classical optimization-based method using shortest-path computation and hill climbing. Second, we formulate the problem as a Markov Decision Process (MDP) and apply a reinforcement learning approach using Q-learning, allowing the agent to learn adaptive decision-making policies.

The main contributions of this work are:

- Modeling the parking assignment problem as both a search problem and an MDP

- Implementing and comparing optimization-based and reinforcement learning approaches
- Evaluating performance under different parking constraints
- Analyzing the trade-off between efficiency and adaptability

The results show that optimization-based methods achieve lower cost in simple scenarios, while reinforcement learning provides greater adaptability in dynamic environments.

2. Related Work

Parking assignment and smart parking systems have been widely studied in intelligent transportation systems. Early work often models the parking problem as an optimization task. For example, Zhao et al.[1] formulated parking planning as a linear assignment problem, where vehicles are assigned to parking spaces based on cost factors such as distance. This approach efficiently utilizes parking resources, but it assumes a relatively static environment and does not account for real-time changes in traffic conditions or parking availability.

Classical optimization techniques are also commonly used for routing problems. Dijkstra [2] in 1959 introduced a shortest-path algorithm for finding optimal routes in weighted graphs. While this method is effective for minimizing travel time, it does not consider additional constraints such as parking cost, availability, or walking distance. In our work, we extend this idea by incorporating these factors into the decision-making process and combining shortest-path computation with a hill-climbing strategy.

More recent research has explored reinforcement learning (RL) for parking and transportation problems. Guo et al.[3] proposed a reinforcement learning-based parking system that models the problem as a Markov Decision Process (MDP) and optimizes long-term decision-making. This approach improves adaptability and resource utilization, but it requires significant training and may face convergence challenges. Similarly, Zhang et al.[4] introduced a multi-agent deep reinforcement learning framework for dynamic parking allocation, which can handle uncertainty in parking demand. However, such methods are computationally expensive and do not always guarantee optimal solutions.

In addition to reinforcement learning, some smart parking systems use deep learning techniques for real-time parking prediction. These systems apply computer vision and time-series models to detect vehicles and estimate parking availability. While these approaches improve real-time awareness, they focus mainly on prediction rather than making optimal parking decisions.

In this project, we build on these ideas by combining classical optimization methods with reinforcement learning. Unlike prior work that focuses on a single approach, we directly compare a graph-based optimization method with a Q-learning-based method within the same framework. This allows us to analyze the trade-off between efficiency and adaptability in the parking assignment problem.

3. System Model and Problem Description

The parking assignment problem is modeled as an intelligent decision-making system in a dynamic urban environment. The system consists of an agent (driver) interacting with a road network environment, where the goal is to select an optimal parking location while minimizing total cost.

The environment is represented as a directed graph, where nodes correspond to locations such as intersections, starting points, and parking lots, and edges represent roads connecting these locations. Each edge is associated with a travel time that may vary depending on traffic conditions. Parking lots are characterized by attributes such as cost, availability, and walking distance to the final destination.

The agent receives input from the environment through sensors, including its current position, travel times, parking availability, and associated costs. Based on this information, the agent selects actions such as moving to adjacent nodes or choosing to park at a specific location. The objective is to minimize a performance measure that includes total driving time, parking cost, and walking distance penalties.

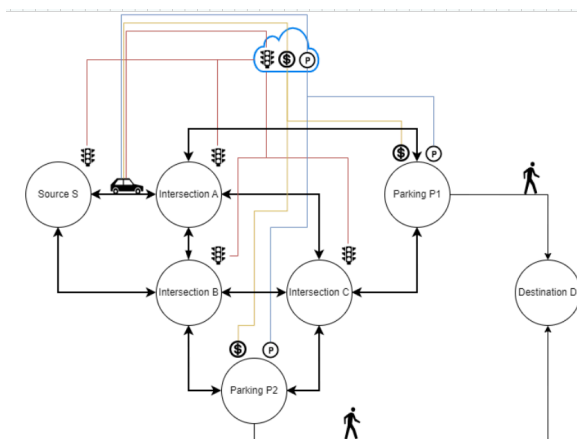


Figure 1: Graph-based system model for the parking assignment problem, including intersections, parking lots, and the agent's path from source to destination.

4. Proposed Approach

To solve the parking assignment problem, we implemented and compared two different approaches: a classical optimization-based method and a reinforcement learning method. The first approach combines shortest-path computation with a hill-climbing optimization strategy, while the second approach models the problem as a Markov Decision Process (MDP) and applies Q-learning. These approaches were selected to evaluate the trade-off between deterministic optimization methods and adaptive learning-based methods in dynamic environments.

4.1 Graph-Based Optimization with Hill Climbing

In the first approach, the road network is modeled as a directed graph where nodes represent locations and edges represent travel paths with associated time costs. To estimate driving cost from the source node to each parking lot, Dijkstra's algorithm is used to compute shortest-path distances.

An objective function is then defined to evaluate each parking lot based on driving time, parking cost, and walking distance. Infeasible options, such as unavailable parking or lots exceeding the walking constraint, are heavily penalized.

Hill climbing is applied to select the optimal parking lot by iteratively comparing candidate solutions and moving toward lower-cost options until no further improvement is possible. This results in a locally optimal solution.

4.2 Reinforcement Learning with Q-Learning

In the second approach, the parking assignment problem is formulated as a Markov Decision Process (MDP), where states represent the agent's location in the road network and actions correspond to movements between nodes or parking decisions.

We apply Q-learning, a model-free reinforcement learning algorithm, to learn an optimal policy through interaction with the environment. A Q-table is used to store state-action values, and actions are selected using an epsilon-greedy strategy to balance exploration and exploitation.

The reward function reflects the optimization objective: movement between nodes incurs a penalty proportional to travel time, while parking rewards are based on parking cost and walking distance. Infeasible actions are heavily penalized to enforce constraints.

Q-values are updated iteratively based on observed rewards and estimated future returns, allowing the agent to learn a policy that minimizes total cost over time. This approach enables adaptation to dynamic and uncertain parking conditions.

4.3 Algorithm Selection Rationale

The two approaches were selected to compare classical optimization methods with reinforcement learning in dynamic environments. The graph-based method with hill climbing provides a deterministic and computationally efficient baseline, while

Q-learning enables adaptive decision-making through interaction with the environment. This comparison highlights the trade-off between solution efficiency and adaptability.

5. Performance Evaluation

To evaluate the proposed methods, we conducted simulation experiments on a graph-based road network consisting of a start node, intermediate nodes, and two parking lots. The goal was to compare the optimization-based approach (Dijkstra + hill climbing) with the reinforcement learning parking approach (Q-learning) under different conditions.

Three scenarios were considered: (1) a baseline case where both parking lots are feasible, (2) a scenario where parking lot P2 exceeds the walking-distance constraint, and (3) a scenario where parking lot P1 is unavailable. These scenarios test the ability of each method to adapt to changing feasibility constraints.

The evaluation considers multiple factors, including the selected parking lot, parking cost, walking distance, and total cost, where total cost combines driving time, parking cost, and walking-distance penalties.

In the baseline scenario, both methods selected parking lot P2, which has a lower parking cost compared to P1. The optimization-based method achieved a slightly lower total cost (14.5) compared to Q-learning (15.5), as it computes the shortest path deterministically using Dijkstra's algorithm. In contrast, the Q-learning approach relies on learned policies, which may result in slightly

suboptimal decisions due to exploration during training.

When parking lot P2 became infeasible due to the walking-distance constraint, both methods correctly adapted by selecting parking lot P1, resulting in a higher total cost (19.5). Similarly, when parking lot P1 was unavailable, both methods selected P2, demonstrating consistent handling of availability constraints.

Table 1 summarizes the performance across all scenarios. In each case, both methods selected the same feasible parking lot, indicating correct constraint handling. However, the optimization-based approach consistently achieved slightly lower cost in the baseline scenario due to its use of shortest-path computation, while Q-learning prioritizes adaptability over strict optimality.

Scenario	Method	Selected lot	Parking cost	Walk dist	Total cost
Baseline	Opt	P2	3	450	14.5
Baseline	RL	P2	3	450	15.5
P2 Too far	Opt	P1	8	350	19.5
P2 too far	RL	P1	8	350	19.5
P1 Unavailable	Opt	P2	3	450	14.5
P1 Unavailable	RL	P2	3	450	15.5

Table 1: Performance comparison between Opt and RL

Figure 2 illustrates the learning behavior of the Q-learning agent, showing that rewards stabilize over training episodes, indicating convergence. Figure 3 compares the final cost across scenarios, highlighting the increase in cost when preferred parking options become infeasible.

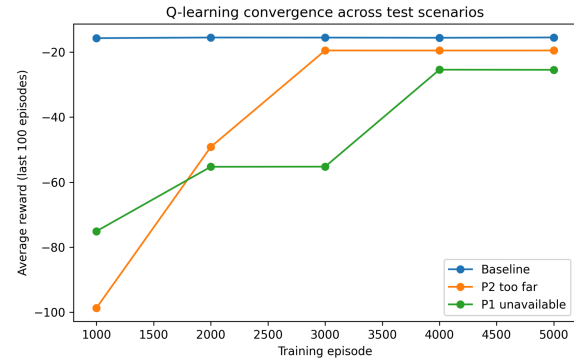


Figure 2: Average reward of the Q-learning agent during training for the three parking scenarios.

Figure 3 compares the total cost across scenarios, showing that cost increases when preferred parking options become infeasible due to walking-distance or availability constraints.

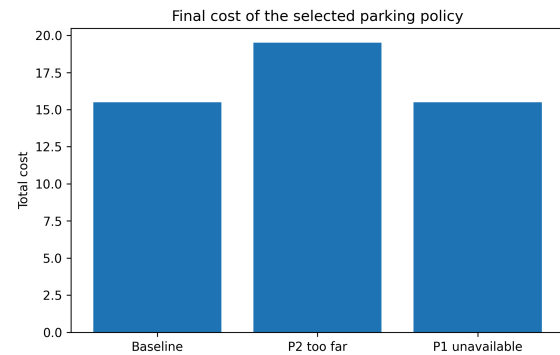


Figure 3: Cost comparison across parking scenarios.

Overall, the results demonstrate that the optimization-based method performs efficiently in static scenarios, while Q-learning provides greater adaptability in dynamic environments. This highlights the trade-off between efficiency and adaptability in solving the parking assignment problem.

6. Discussion and Conclusion

In this project, we compared a classical optimization-based approach with a reinforcement learning approach for the

parking assignment problem. Both methods successfully selected feasible parking locations across all test scenarios, demonstrating correct handling of constraints such as availability and walking distance.

The optimization-based method, which combines Dijkstra's algorithm with hill climbing, achieved slightly lower cost in the baseline scenario due to its use of shortest-path computation. However, the Q-learning approach demonstrated the ability to learn adaptive decision-making policies through interaction with the environment, making it more suitable for dynamic and uncertain conditions.

These results highlight a key trade-off between efficiency and adaptability. While

traditional optimization methods perform well in static environments, reinforcement learning offers greater flexibility when conditions change over time.

Future work could extend this model to larger and more complex road networks, incorporate real-time traffic data, or explore advanced reinforcement learning techniques such as deep Q-learning to improve scalability and performance.

These findings suggest that reinforcement learning is a promising direction for intelligent parking systems operating under uncertainty, while classical optimization remains effective when the environment is stable.

References

- [1] X. Zhao, K. Zhao, and L. Zhang, “Parking Planning as a Linear Assignment Problem,” arXiv preprint arXiv:1708.07932, 2017.
- [2] E. W. Dijkstra, “A note on two problems in connexion with graphs,” *Numerische Mathematik*, vol. 1, pp. 269–271, 1959.
- [3] Y. Guo, et al., “Reinforcement Learning-Based Parking System Optimization,” *Mathematical Biosciences and Engineering*, 2024.
- [4] Z. Zhang, et al., “Multi-Agent Deep Reinforcement Learning for Dynamic Parking Allocation,” *Applied Soft Computing*, 2022.